



cPRC

$$\frac{dC_P(t)}{dt} = \overset{\text{Baseline/ steady state Ca}^{2+} \text{ concentration}}{B(\textcolor{brown}{S}, \textcolor{blue}{N}_{in,2}, t)} - \delta_{C_P} C_P(t) + \overset{\text{late NO dependent increase}}{\kappa_{C_P} N_{in,2}(N, t)} + \overset{\text{UV dependent dynamics}}{UV_{in}(UV, \textcolor{brown}{S}, N_{in,1}, t)}$$

Ca²⁺ dynamics (**measured**)
dependent on UV, S(UV,t),
N_{in,1}(N,t) and N_{in,2}(N,t)

NIT-GC1 dependent NO uptake
that acts on Ca²⁺ dynamics via:
N_{in,1}(N,t) and N_{in,2}(N,t) (late)

S(UV,t)
NIT-GC2 and UV dependent
phototransduction

NO input to cPRC

NIT-GC2-dependent
input to IN^{NOS}

IN^{RGW}

$$\frac{dC_R(t)}{dt} = \frac{A(C_P, t)}{1 + I(C_N, t)} - \delta_{C_R} C_R(t)$$

Ca²⁺ dynamics (**measured, not modelled**)
dependent on input from cPRC and IN^{NOS}

N(C_N, t)
NO production (**measured**)
dependent on IN^{NOS} Ca²⁺

$$\frac{dC_N(t)}{dt} = \kappa_{C_N} \textcolor{brown}{S}(UV, t) - \delta_{C_N} C_N(t)$$

Ca²⁺ dynamics (**measured**)
dependent on S(UV,t) input from cPRC

IN^{NOS}